

2-The Internet, the Web, and Electronic Commerce

Introduction



- The Internet
 - Interconnected mesh of computers and data lines
 - Connecting millions of people and organizations
- Foundation of the digital revolution
- The web
 - Provides an easy-to-use interface to Internet resources
- To be effective and efficient
 - End user awareness of the resources available on the Internet and the web

The Internet and the Web



- The Internet
 - Large global network connecting smaller networks all over the globe
- The Internet launched in 1969
 - The United States funded a project to develop a national computer network
 - ARPANET (Advanced Research Project Agency Network)
- World Wide Web or WWW was introduced in 1991
- The Internet and the web are NOT the same
 - The Internet is the physical network
 - The web is a multimedia interface to the resources available on the Internet



Web

- Web 1.0
 - 1st generation
 - Linking existing information focus
 - Search programs were created
 - Provide links to websites with specific words or phrases
- Web 2.0
 - Dynamic content creation
 - Facebook is most common in Web 2.0
- Web 3.0
 - Identifies relationships between data
 - Personalized content creation for users

Common Internet Uses

- Communicating
 - E-mail, photos, videos
 - Discussions
- Shopping
- Searching
 - Virtual libraries
- Education or e-learning
- Online Entertainment

Making IT Work for You
Online Entertainment

- Online options for TV, movies, music, books, social media, news feeds
 - Hulu, Amazon Prime, Netflix, Amazon Kindle, CNN, Facebook, Twitter

Internet Access Providers

- Internet Service Provider (ISP)
 - Common way to access the Internet
 - Provide a path to access the Internet
 - Use telephone lines, cable, and/or wireless connections
- Most common providers
 - Verizon, Comcast, Sprint, T-Mobile, AT&T

Browsers

- Provide access to web resources
- Allow you to explore the web
 - Connect to remote computers
 - Uncomplicated interface to the Internet
 - File transfer
 - Display many varieties of multimedia
- Popular Web browsers include:
 - Mozilla Firefox
 - Apple Safari
 - Microsoft Edge
 - Google Chrome

URLs

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- Uniform Resource Locator
 - Location or address of resource
 - https is the most common for web traffic
 - Two parts
 - Protocol
 - Rules for exchanging data
 - Domain name
 - Where resource is located

Top-Level Domain (TLD)

Top-level domain (TLD) or Web Suffix

- Identifies the type of organization
 - .com Commercial
 - .edu Educational
 - .gov Government
 - .mil U.S. military
 - .net Network
 - .org Organization
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HTML and Hyperlinks

- Hypertext Markup Language
 - Markup language for displaying web pages
- Browsers interpret HTML commands
 - Display document as a web page
- Hyperlinks or links
 - Connect to other web pages
 - Text files
 - Graphic images
 - Audio and Video Clips

Interactive Web Sites

- Technologies used to provide highly interactive and animated websites
 - Cascading Style Sheets (CSS)
 - JavaScript
 - PHP
- Mobile Browsers
 - Designed to run on portable devices

Web Utilities

- Specialized utility programs that make using the Internet and web safer and easier
 - Filters
 - File Transfer Utilities
 - Internet Security Suites

Filters

- Block access to selected sites
- Set time limits
- Monitor total time spent on the Internet and at individual web sites
 - Net Nanny
 - Qustodio Parental Control
 - Circle with Disney
 - Norton Online Family
 - McAfee Family Protection

File Transfer Utilities

- Upload and download files to and from the Internet
 - Downloading
 - Uploading
- Three popular types of programs
 - Web-based file transfer services
 - Bit-Torrent
 - File transfer protocol (FTP) / Secure file transfer protocol (SFTP)

Internet Security Suites

- Designed to maintain your security and privacy while on the web
- Two best known suites
 - McAfee Internet Security
 - Symantec Norton Internet Security

Communication

- Communication is the most popular Internet activity
 - Social Networking
 - Blogs and Microblogs
 - Webcasts, Podcasts, and Wikis
 - E-mail and Messaging

Social Networking

- Connecting people and organizations that share a common interest or activity
 - Common features
 - Profiles
 - Pages
 - Groups
 - Friends
 - News feed
 - Share settings

Blogs, Microblogs

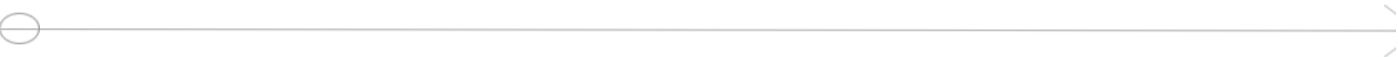
- Blogs
 - Personal websites
 - Date/time-stamped
 - Arranged with the most recent items shown first
- Microblogs
 - Short status updates
 - Most common is Twitter
 - Tweets are Twitter messages

Webcasts, Podcasts, Wikis



- Webcast
 - Streaming technology for live broadcast of audio and video
 - No files after streaming ends
- Podcast
 - Must download files to use
 - Can transfer to media player
- Wiki
 - Specially designed Web site
 - Allows visitors to edit the contents
 - Supports collaborative writing

E-Mail



- E-mail
 - Transmission of electronic messages over the Internet
 - Three elements
 - Header
 - Address
 - Subject
 - Attachments
 - Message
 - Signature

E-mail Systems

- Client-based
 - E-mail client
 - Must be installed on computer
 - Apple Mail
 - Microsoft Outlook
- Web-based
 - Webmail client
 - No installation necessary -- free
 - Google's Gmail
 - Microsoft's Hotmail
 - Yahoo!'s Yahoo!mail

E-mail

- SPAM
 - Unwelcome-mail
- Computer viruses
 - Destructive programs
 - Attached to unsolicited email
- CAN-SPAM Act
 - Created antispam laws for control
 - Every marketing related e-mail must provide an opt-out option
- Spam blockers / spam filters
 - Identify and control spam

Messaging

- Text messaging / texting
 - SMS (short message service)
 - Short electronic message
- Multimedia Messaging Service (MMS)
 - Send
 - Images
 - Videos
 - Sounds

Instant Messaging

- Instant messaging (IM)
 - Extension of email that provides direct, live communication between two or more people
 - Most programs include video conferencing features, file sharing and remote assistance
 - Facebook Messenger
 - Google Hangouts

Search Tools

- Search Services
 - Operate websites
 - Spiders
 - Look for new information and update websites
- Search Engines
 - Assist in locating specific information
- Specialized Search Engines
 - Focus on subject specific websites

Content Evaluation

- Content Evaluation
 - Not everything on the Internet is accurate
 - Authority
 - Accuracy
 - Objectivity
 - Currency

Electronic Commerce

- E-commerce
 - Buying and selling of goods over the Internet
 - Three basic types
 - Business-to-consumer (B2C)
 - Consumer-to consumer (C2C)
 - Business-to-business (B2B)

Business to Consumer (B2C)

- Sale of product or service to general public
- Fastest growing type of e-commerce
- Three most widely used B2C applications:
 - Online banking
 - Financial trading
 - Shopping
 - Amazon.com is one of the most widely use B2C sites

Consumer to Consumer C2C

- Consumer-to-consumer e-commerce (C2C)
 - Individual to individual
- Web auctions
 - Buyers and sellers seldom meet
 - Bids are submitted electronically
 - Person-to-person auction sites

Business to Business B2B

- Sale of a product or service from one business to another
- Primarily a manufacturer supplier relationship

Security

- Payment methods must be fast, reliable, and secure
 - Provide a convenient way to submit buyers information
- Two options
 - Credit card
 - Fast and convenient
 - Digital cash
 - Internet's equivalent to traditional cash
 - Converts digital cash to currency through 3rd party

Cloud Computing

- Shifts computing activities from users' computers to computers on the Internet
- Frees end-users from owning, maintaining, and storing software programs and data
- Three basic components:
 - Clients (end-users)
 - The Internet
 - Service providers

Internet of Things (IoT)

- Continuing development of the Internet
- Allows objects embedded with electronic devices to send and receive data
 - Smartphones
 - Wearable devices
- Uses Web 3.0 Applications

Careers In IT - Webmasters



- Develop and maintain websites and resources
 - Backup of company website
 - Design and development of websites
 - Work with the marketing
 - Increase site traffic
 - Development of web promotions

A Look to the Future

- Car's Dashboard as a power, Internet-connected computing device
 - Safety and diagnostic features are already there
- Self-driving cars